

Implementation of the data science aspect for the U-Book application

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Abstract—The discovery of truth, also known as the pursuit of veracity, has become one of the fundamental needs in Big Data, which is the 4th V. We have developed a web application called "U-Book" for the exchange of educational documents. This exchange ensures the accuracy of information by verifying the metadata of these documents, specifically their disciplines, authors, and years of establishment. This information is verified through a machine learning model based on a set of decision trees. "U-Book" is also designed for the management of university events, ensuring the accuracy of their execution and the registration of participants. To achieve this, we have adopted a veracity model using web scraping.

Index Terms—U-Book, Veracity of educational documents, Veracity of university events, Machine learning, Big Data, Web Application

Additionally, we will concentrate on the last two aspects by defining them, mentioning the different types, and models.

I. PROJECT OVERVIEW

First, we will define the environment in which this project took place by introducing the "U-Book" Startup. Then, we will briefly describe the issue encountered in this context, highlighting the proposed solution. Finally, we will conclude by presenting the overall report structure focusing on the Data Science side.

The application was designed and developed using Angular for the front end, Spring for the back end and Python for the Data Science proposition.

II. PROBLEMATIC

Statistics (Customer Discovery, Questionnaires, Research) during the Carthage Innov1 2021 entrepreneurial competition indicate that:

- There is a lack of event management spaces in Tunisia, specifically within the university context.
- There is a absence of platforms and portals that offer a free service for accessing educational documents.
- There is a lack of credibility and veracity of educational documents on the internet, making it difficult to find relevant documents that meet the needs.
- There is no Tunisian system that handles mock exams and issues knowledge badges.

III. PROPOSED SOLUTION

This problem led to the idea of developing an educational university encyclopedia that provides various educational resources (lectures, tutorials, practical work, exams, answers) with the possibility to take mock exams to earn HardSkills or SoftSkills competence badges, as well as an online events center (training, certifications, competitions, and special days).

And from the above, the name U-Book was derived:



Fig. 1: U-Book brand

INTRODUCTION

U-Book is a social network that offers student life services. It features an event management space and a platform for sharing educational documents. One of the significant challenges is ensuring that the platform adheres to ethical guidelines, particularly the veracity of information published regarding educational documents and university events.

In this paper, we will discuss events, define them, list their different types, and frequently used policies. Then, we will delve into educational documents, specifically identifying discrepancies between the types of educational documents and their metadata. We will also present an example of one of the policies for documents, such as Creative Commons licenses.

After this focus on the accuracy of information, we will have a better understanding of the concept and its influence on documents and events. To achieve this aspect effectively, we will turn to artificial intelligence, which will, in turn, use indexing as a tool to accomplish the task.



Fig. 2: U-Book Explanation

- U stands for University.
- BooK corresponds to the English word "book" or "document".
- U-BooK represents university documents (university books).
- BooK is also a verb in English for making reservations.
- U-BooK can also be defined as "you book up in events," meaning you make reservations for events.

IV. OBJECTIVES OF U-BOOK

- Facilitate the search for educational resources.
- Improve the sharing of educational documents while adhering to ethical document exchanges.
- Increase event interactions.
- Expand the university network.

V. STATE OF ART

A. Incorporation of Sentences (SIF)

Recent research blends word embeddings through vector and matrix operations, revealing the efficacy of vector exploitation via coordinate-wise multiplication in binary operations. This study emphasizes distributed sentence representations, typically necessitating syntactic analysis. An alternative method employs an unsupervised algorithm for acquiring distributed sentence or document representations, demonstrating competitive performance relative to contemporary methods [1].

B. Syntax Analysis

Syntax analysis is a foundational technology in natural language processing and a cornerstone for a profound understanding of language. The goal of syntax analysis is to identify the syntactic components within a sentence and the relationships between these components, typically represented using parse trees to convey the results of the syntax analysis [2].

C. Calculation of sentence embedding by merging the syntactic parse tree and word embedding

The project introduces a sentence embedding method using syntactic tree analysis and word vectors. Node fusion in the syntactic tree preserves word order. Yong Wang and Shuixiu Wu consider syntactic tree tags as weight parameters, enhancing sentence embedding with syntactic information. The method shows potential in overcoming existing weaknesses, offering fast computation, easy parameter learning, and competitive performance compared to traditional SIF in text similarity tasks. However, there are areas for improvement, including optimizing parameter learning and designing an unsupervised learning target for enhanced performance[3].

VI. EDUCATIONAL DOCUMENT VERACITY

A. Model Presentation

To validate the authenticity of an educational document, we analyze the metadata more specifically, including discipline, author, and document year. In U-BooK,

the user will input fields such as Subject, Year, and a Creative Commons license that contains information about the document's author, along with the document itself.

We will expedite the extraction of information about the year and the author by parsing the documents, using tools that read the metadata extension of the document in PDF, DOCX, or PPTX format.

Regarding the document's discipline, we are developing a machine learning model to verify it. To create this model, we require a dataset to train it using a learning algorithm.

Finally, we compare the fields entered by U-BooK users with the values obtained through document scraping. If they match, we assume that the validity is upheld.

This figure 3 try to summarize the process for the U-BooK logic.

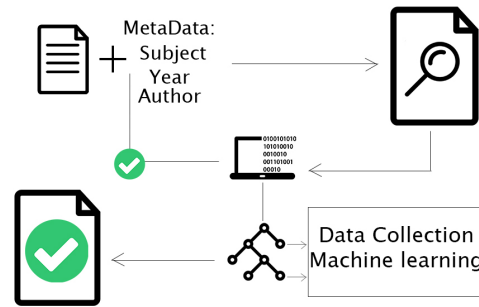


Fig. 3: Process of verifying the truthfulness of educational documents

B. Data Set

Since U-BooK is still in the startup phase, we choose broad subjects as the dataset categories:

- Health Sciences: Encompassing everything related to medicine, pharmacy, nursing, anesthesia, etc.
- Life Sciences (SVT): Covering topics in life sciences, biology, geology, earth sciences, and more.
- Chemistry: Including organic chemistry, agrochemicals, and related fields.
- Physics: Encompassing mechanics, electricity, astronomy, and other physical sciences.
- Mathematics: Covering topics like analysis, algebra, probability, and more.
- Computer Science: Involving development, networking, telecommunications, and related areas.
- Literature: Encompassing languages such as French, English, Arabic, and related subjects.
- Management: Including finance, economics, taxation, and other management-related topics.
- Other: Reserved for subjects like music, art, and other categories that may be added in the future.

For each section, it's possible to further divide them into subsections as the application progresses, creating a subtle hierarchical structure.

The data to be collected includes various types of documents (lectures, labs, tutorials, summaries, exams, quizzes) across multiple disciplines. These documents will be grouped according to the decision tree structure presented in the model overview in the same chapter.

You mentioned that you initially looked into Kaggle but didn't find the specific documents you were searching for. In response, you reached out to students from various fields to provide their educational documents. Additionally, you conducted some internet research. All these efforts led to the creation of your own dataset, as shown in Figure 4.

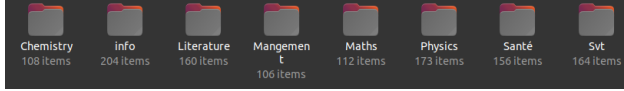


Fig. 4: DataSet de U-Book

This custom dataset will serve as the foundation for U-Book's document classification and organization according to the decision tree structure.

C. Machine Learning Model

We are now at the stage of classifying documents into the categories mentioned in the previous section.

Popular classification algorithms include K-Nearest Neighbors (KNN), Naïve Bayes, Decision Trees, and Decision Forests.

KNN (K-Nearest Neighbors) has been ruled out because it consumes too many computational and hardware resources, especially with a detailed dataset. According to [60], the complexity in KNN is of the order $O(n^2)$ with $k \ll n$, where n is not the number of documents in the entire dataset. Therefore, with 9 classes ($k = 9$), and assuming there are at least 30 documents for each class, $n = k * 30 = 9 * 30 = 270$, resulting in exponential complexity.

Regarding the Naïve Bayes algorithm, it assumes that each feature is independent, which is not always the case. In our scenario, there can be documents that relate to two subjects, such as mechanics and electronics in a physics course.

As for the decision tree algorithm, it works well when the classification is binary (true or false), but it often gives poor results when there are multiple answers. Therefore, the decision tree set is chosen because it can handle more than just binary questions, and decision trees are easy to interpret visually.

In the next step, we choose between the decision tree and the decision forest, which is an ensemble of decision trees. The decision forest is composed of multiple decision trees, as shown in Figure 5.

This decision forest approach allows for more robust and accurate classification of documents in U-Book.

From the data collected, we reserve 20% of the data for testing, and the data distribution will be random within each subclass. For example, in the "Computer

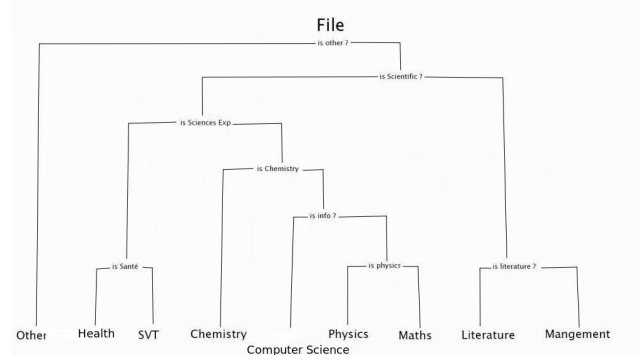


Fig. 5: U-Book Decision Trees

Science" class, we have divided the documents into 80% for training and 20% for validation testing.

VII. PSEUDO-CODE FOR DOCUMENT SUBJECT CLASSIFICATION

A. Text Extraction

```

1  \\ Import libraries for PDF, Word,
   PowerPoint Document
2  import PDF
3  import DOCX
4  import PPTX
5
6  \\ Define variables
7
8  var File = "" \\ this is the link for the
   file we're treating
9  var txt = "" \\ this is the result of the
   txt inside the document
10
11 \\ Main Method
12
13 Method init(documentLink) {
14     this.file = documentLink
15 }
16
17 \\ Method for determining the file either
   PDF, DOCX or PPTX
18
19 Method detectFile (file) {
20     if (file.endsWith('.pdf')) {
21         txtFromPdf(file)
22     } if (file.endsWith('.docx')) {
23         txtFromDocx(file)
24     } if (file.endsWith('.pptx')) {
25         txtFromPptx(file)
26     }
27 }
28
29 \\ Method to Extract text from PDF
30
31 Method txtFromPdf(file) {
32     pdf = PDF(file)
33     for i in range (pdf.numberOfPages) {
34         page = pdf.getPage(i)
35         txt += page.getText()
36     }
37 }
38
39 \\ Method to Extract text from DOCX
40
41 Method txtFromDocx(file) {
42     doc = DOCX(file)
43     for i in range (doc.

```

```

        numberOfParagraphes) {
44         paragraphe = doc.getParagraphe(i)
45         txt += paragraphe.getText()
46     }
47 }
48
49 \\ Method to Extract text from PPTX
50
51 Method txtFromPptx(file) {
52     ppt = PPTX(file)
53     for i in range (ppt.numberOfSlides) {
54         slide = ppt.getSlide(i)
55         txt += slide.getText()
56     }
57 }

```

B. Racinisation and lemmatisation

```

1  \\ Import Libraries
2
3  import EnglishFrenshStopWords
4  import removingAccents
5  import stemming
6  import lemmatisation
7
8  \\Define variables
9
10 var tokens \\ List of tokens
11
12 \\ Method removing extra spaces
13
14 Method RemovingExtraSpaces(txt) {
15     txt = txt.trim() \\Removing spaces
16     from the beginning and the end of the
17     string
18     txt = txt.replace("\\n","") \\Removing
19     back to line from string
20     txt = txt.replace("\\t","") \\Removing
21     tabulations from string
22 }
23
24 \\ Method tokenisation
25
26 Method Tokenization(txt) {
27     tokens = txt.split(" ") \\split text
28     into list with the space as the
29     indicator
30 }
31
32 \\ Method Removing stop words
33
34 Method StopWords(tokens) {
35     var tokens2
36     for i in tokens {
37         if (i not exist in
38             EnglishFrenshStopWords)
39             tokens2.add(SpecialCarac(i
40             ))
41     }
42     tokens = tokens2
43 }
44
45 \\ Method Removing special caracteres
46
47 Method SpecialCarac (token) {
48     return removingAccent(lemmatisatoin(
49     stemming(token)))
50 }

```

C. Document Subject Predetection

```

1
2  \\ Import Librairies
3
4  import MachineLearningVeracitySubjectModel
5
6  \\ Declaring variables
7
8  var tokens = "" \\ list of tokens
9  extracted from the last pseudo code
10 var subject = "" \\ result we're searching
11 for
12
13 \\ Main method
14
15 init(tokens) {
16     this.tokens = tokens
17     model =
18     MachineLearningVeracitySubjectModel.
19     load()
20     subject = model.predict(tokens)
21 }

```

VIII. IMPLEMENTATION AND DEPLOYMENT

In this section, we'll present the final implementation of this model on the web application of U-Book, through the model we can afford a verified badge for the document and an email will be sent for the author of that document, in both case:

- Verified Document: a confirmation mail for the submission
- Unverified Document: asking the author to verify his document, and showing the exceptions of that document

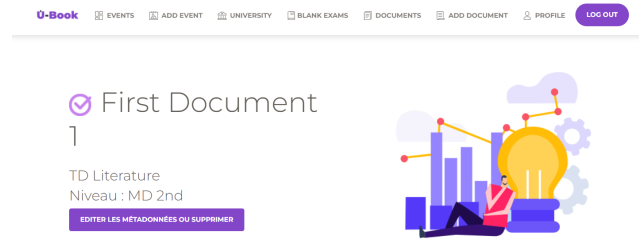


Fig. 6: DataSet de U-Book

IX. UNIVERSITY EVENTS VERACITY

A. Model Presentation

The verification of the authenticity of events is done through two aspects:

- 1) Verification of the credibility of the event's execution.
- 2) Verification of registrations.

Additionally, ensuring the accuracy of events involves verifying that there are no false registrations and false events.

a) *Verification of the credibility of the event's execution*

To ensure the credibility of events on the U-Book platform, two methods are employed:

- 1) Integration with external sources: U-Book eliminates the need for web scraping and instead showcases events from credible external sources like 4C or 10Times. This guarantees the authenticity of events.
- 2) User psychological assessment: U-Book relies on the psychology of users. It is unlikely that someone would pay to publish a fake event. Therefore, if a U-Booker posts an event and is supervised by another U-Booker, such as a club or institute, U-Book verifies the event's authenticity by contacting the moderator via email. If the moderator confirms the event, its authenticity is ensured. However, if the moderator does not confirm or if the U-Booker is independent, an option to pay for event publication on the platform is provided. If the payment is made, the event's authenticity is guaranteed. Otherwise, the event remains hidden until one of the two solutions is realized: either payment or confirmation by the supervisor.

b) *Verification of registrations*

To ensure that most of the registrants attend the event, it is necessary for them to enter their login and password during registration. This approach encourages registrants to take their participation in the event seriously and adds a layer of authentication to their commitment.

We are going to present the web scraping part to retrieve genuine events from official event websites and pages.

We need knowledge about the structure of the page we are scraping. Understanding the structure of the target website or page is crucial for effective web scraping. This includes identifying HTML tags, CSS classes, and other elements that contain the event information you want to extract. The better you understand the page's structure, the more accurately you can retrieve the relevant data. Additionally, you may need to analyze the page's structure and adapt your scraping code if the structure changes over time.

B. *Pseudo-Code For the University Events web scraping*

We are going to present the web scraping part to retrieve serious events from official event websites and pages. We need knowledge of the page structure that we are scraping.

```
1 # Import libraries
2 import outilDeGrattage
3 import RequesHttp
4
5 # This is the link to the page
6 var webpage = ""
7
```

```
8 # This is the HTML content of the page
9 var src = ""
10
11 # This is the scraping result
12 var content = ""
13
14 # List of events you are searching for
15 var events = []
16
17 webpage = RequestHttp.get('website link ')
18 src = webpage.text
19 content = outilDeGrattage(src)
20 events = content.find_all('<div class '+'=
    "event">')
```

X. IMPLEMENTATION

For the implementation of this application, we choose Python (version 3.8.10) as the programming language, along with the following tools:

- Beautiful Soup: A web scraping tool that is fast and easy to use.
- Requests: A tool for sending HTTP requests, which is used to load the web page being scraped.
- mysql.connector: Used to add the results found into the U-Book application's database.
- PyPDF2: A library for processing PDF files.
- python-pptx: A library for processing PPTX files (PowerPoint files).
- python-pptx: A library for processing PPTX files (PowerPoint files).
- NLTK: A library for natural language processing.
- NumPy: A library for managing lists and arrays.
- Pandas: A library for managing data frames.
- joblib: A library for saving and loading trained models.
- scikit-learn (sklearn): A machine learning library for various models.
- os: A library for handling system commands.

The implementation is done using the PyCharm text editor by JetBrains on the Ubuntu v20.04 LTS operating system. This combination of tools and environment is commonly used for web scraping projects in Python.

XI. RESULTS AND DISCUSSION

In this chapter, we analyze the truth-seeking model for educational documents. We used the metrics from the scikit-learn library: MAE (Mean Absolute Error) $= \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$, MSE (Mean Squared Error) $= \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$, RMSE (Root Mean Squared Error) $= \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$, $R2 = 1 - \frac{MSE}{\frac{1}{n} \sum_{i=1}^n (y_i - \bar{y})^2}$, which is a final score derived from the metrics.

We calculated the test score for each decision tree illustrated in the U-Book model chapter for document truthfulness, in the Model Overview section, under the Learning Algorithm subsection.

¹Division by the number of tests.

²The expected value.

³The predicted value

⁴Average of the y values

For final model, we will present the test scores and training scores.

We can assess the model quality from the scores as follows:

- If the score is < 50 , the model quality is poor.
- If the score is in the range $[50..70]$, the model quality is fair.
- If the score is in the range $[70..90]$, the model quality is good.
- If the score is > 90 , the model quality is excellent.

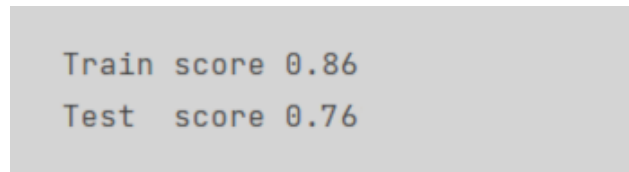


Fig. 7: Average Score For Final Model

We can deduce that the outcomes are encouraging for a data science project from scratch.

CONCLUSION

As part of the end-of-studies project, we participated in the development of a document exchange and event management application while adhering to truthfulness ethics. This report has presented in detail all the steps we went through to achieve the expected result.

As for the truthfulness part, it is currently being improved and will be represented on different levels of truthfulness. The service seeking this truthfulness will be deployed on a Flask server, for example, and will communicate with the application through web services.

Furthermore, we will design a model that verifies the document type and other metadata to ensure the truthfulness of educational resources.

Finally, the application's design is still ongoing, and improvements are being made through explanatory videos on how the application functions.

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